

Charles Batsaikhan

511 Lexington Parkway S, #5, Saint Paul, MN, 55105
(952) 923-9267 | cbatsaik@macalester.edu | [Portfolio](#) | [LinkedIn](#)

EDUCATION

Macalester College

St. Paul, MN – May 2025

Bachelor of Arts – Statistics, Minor - Mathematics

GPA: 3.65

Honors: Macalester College Dean's List (Spring 2024, Fall 2025)

Relevant Coursework: Statistical Modeling, Statistical Machine Learning, Intermediate Data Science, Database Management Systems, Computational Linear Algebra, Causal Inference, Probability, Statistical Theory, Object Oriented Programming

SKILLS

Programming Languages: R, Python, SQL, Java

Tools/Frameworks: tidyverse, pandas, NumPy, TensorFlow, Shiny, Power BI, ArcGIS, DBI, ODBC, dbplyr, Excel

EXPERIENCE

Data Scientist Intern, Transit Information

Minneapolis, MN

Metro Transit (Metropolitan Council)

June 2025 – Present

- Build and maintain an **R Shiny** tool that tests ridership thresholds and compares 2019 vs 2024; prioritize **2,130** stops for **shelter upgrades**, improving service for **~4,200** daily riders and informing Better Bus Stops policy.
- **Engineer an R/SQL** pipeline unifying schedules, sensor-based ridership, route paths, and shelter inventory; **standardize metrics**, add validations, and enable reliable self-serve analysis.
- **Model Real-Time Sign** data and ship a drill-down app (sign -> issue history); **reduce refresh failures 30%** and speed maintenance triage.
- Document lineage/validation (Quarto), implement Git/PR reviews, and write reusable dbplyr aggregators (by route/direction and by stop) to ensure consistent reporting.

Teaching Assistant

Saint Paul, MN

Macalester College (Intro to Data Science, Statistical Machine Learning, Probability)

Jan 2024 – May 2025

- Led labs/office hours; taught R workflows (**dplyr/ggplot2**), ML methods (**LASSO, logistic regression, decision trees**), and probability (**distributions, CLT, MLE**).
- Created review guides and example notebooks; graded with rubrics and provided targeted feedback on code, interpretation, and reproducibility.
- Ran debugging sessions and mini demos in **R/tidyverse**; emphasized data hygiene, EDA, model evaluation, and clear communication of results.

Kokua Kalihi Valley

Honolulu, HI

Summer Accounting Intern

May 2022 - Aug 2022

- Processed and posted journal entries in **MIP Fund Accounting**; reconciled accounts and resolved discrepancies with supporting documentation.
- Built **monthly expense summaries** in Excel (pivot tables, LOOKUPS, templates) to streamline reporting for program managers.

PROJECTS

Capstone – [Analyzing the Environmental Legacy of Racial Covenants in Minneapolis](#)

Project Team Lead

- Used **causal inference** and **spatial analysis** (R, ArcGIS Pro) to examine links between 1920s covenants and current environmental inequality.
- Integrated air quality, Social Vulnerability Index, and land-surface temperature into an **Environmental Injustice Score**; presented policy-relevant findings.

Opioid Epidemic – Intermediate Data Science course

- Cleaned and processed large datasets using R (Tidyverse) for data-driven insights.
- Applied **Random Forest** models to identify key predictors of opioid overdose death.
- Created RShiny dashboard and regional maps for trend exploration and spatial analysis.

Powerlifting Outcome – Statistical Machine Learning

- Predicted competition outcomes (win/qualify) to inform attempt selection and training focus.
- Trained **LASSO** and **logistics regression** with time-based cross-validation and hyperparameter tuning.
- Improved AUC **from 0.71 to 0.84**, boosted **F1 by 18%**, and reduced **log-loss by 22%**; explained drivers via coefficient paths and partial effects.

LEADERSHIP ACTIVITIES

Macalester Men's Swimming and Diving Team

September 2021 – May 2025

- Captain of the swim and dive team.
- Nominated Team "MVP" for four years straight.
- Won Gold, Silver, Bronze Medals in the Minnesota Intercollegiate Conference Championship.

